

Hybrid Cloud Architecture Design

Global Retail Platform — Hybrid Cloud Modernization Initiative

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1. Architecture Overview

The Global Retail Platform adopts a hybrid cloud architecture that distributes workloads across two primary environments: a set of regional on-premises data centers and IBM Cloud infrastructure spanning the US-South (Dallas) and EU-Central (Frankfurt) regions. This design preserves existing investments in on-premises systems where regulatory, operational, or integration constraints require it, while migrating customer-facing and scalability-critical workloads to the cloud to meet peak demand targets of 50,000 concurrent users.

The architecture is designed around five foundational principles drawn directly from the stakeholder and requirements analysis:

- Workload sovereignty — regulated data (EU PII, payment tokens) never leaves defined compliance boundaries
- Elastic scale — cloud-native services on IBM Kubernetes Service (IKS) scale horizontally in response to real-time demand
- Resilience by design — multi-zone IKS clusters, cross-region replication, and automated failover target 99.95% availability
- Security in depth — network segmentation, mTLS service mesh, encryption at rest and in transit enforced at every tier
- Unified operations — IBM Satellite and IBM Cloud Toolchain manage hybrid and cloud workloads through a single control plane

1.1 Environment Summary

On-Premises Data Centers (US & EU)

- POS terminals, ERP, Inventory WMS, and legacy order core — systems with deep hardware or financial integrations
- EU customer PII database and payment token HSM vault — GDPR and PCI-DSS mandated on-prem or EU-region residency
- Connected to IBM Cloud via IBM Cloud Direct Link 2.0 (primary) and IPSec VPN (failover)

IBM Cloud — US-South (Dallas) — Primary Production Region

- All customer-facing microservices: Product, Order Processing, Checkout & Payment, Customer Profile, API Gateway

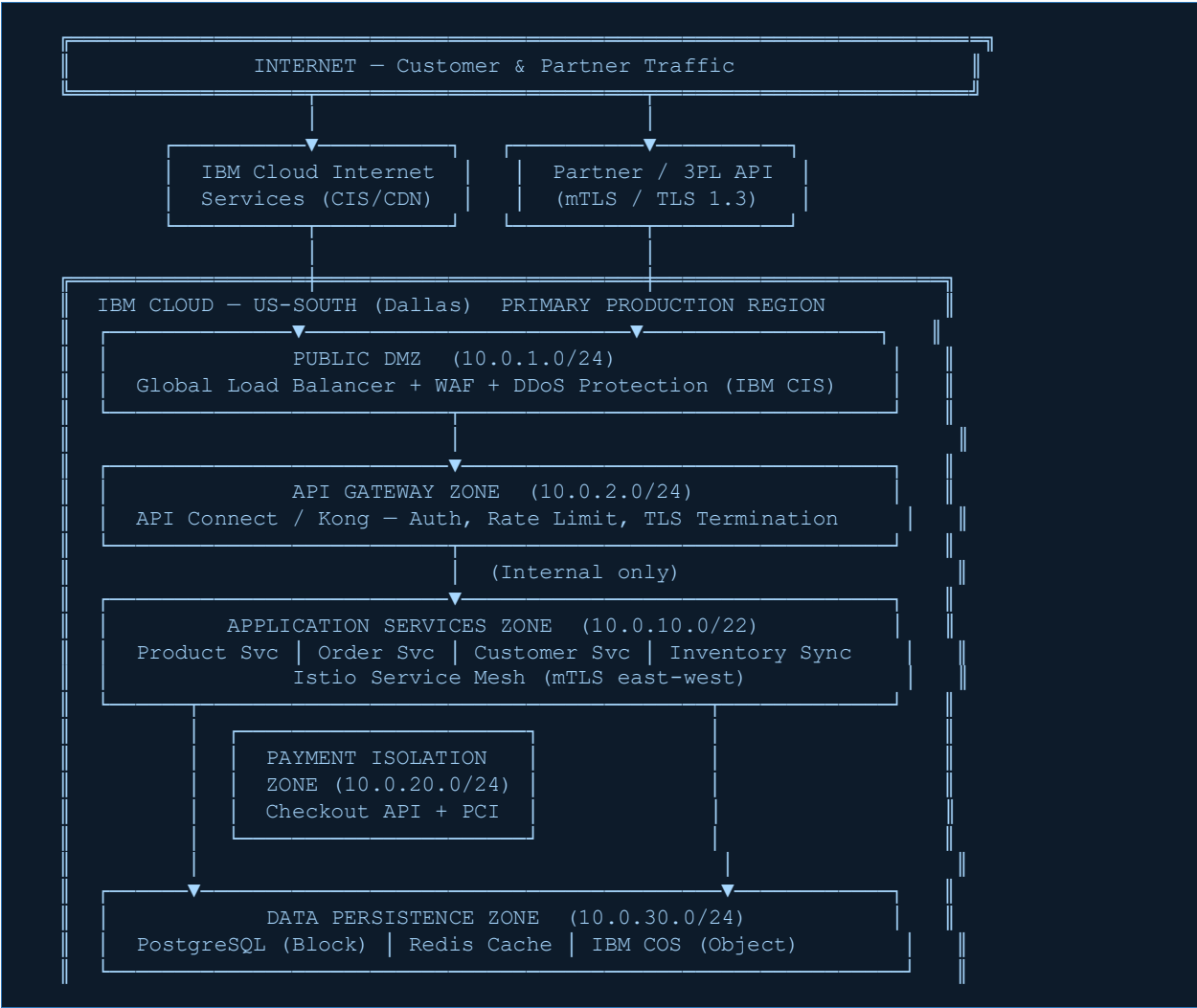
- IBM IKS multi-zone worker pools; IBM Cloud Databases (PostgreSQL, Redis); IBM Cloud Object Storage
- CI/CD pipeline (IBM Cloud Toolchain); Observability platform (IBM Cloud Monitoring + Log Analysis)

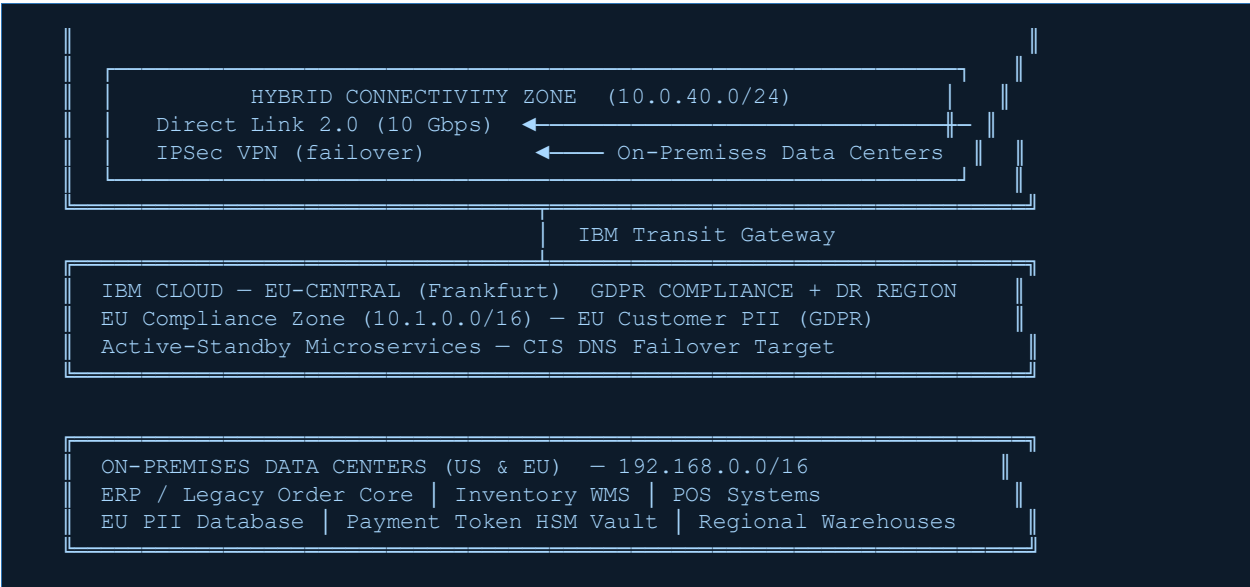
IBM Cloud — EU-Central (Frankfurt) — GDPR Compliance Region

- EU Customer Profile Database — data residency enforced; no cross-region replication of PII
- Active-standby deployment of core microservices for DR purposes
- Disaster recovery target: automated CIS DNS failover within 60 seconds of US-South region health check failure

2. High-Level Architecture Diagram

The diagram below illustrates the end-to-end hybrid cloud topology, showing the relationship between on-premises infrastructure, IBM Cloud regions, and external integration partners.





3. Workload Placement

Workload placement decisions are driven by four criteria: regulatory compliance requirements, dependency on legacy infrastructure, horizontal scalability needs, and latency sensitivity. The table below documents the placement rationale for every major platform component.

Component	Environment	Placement	Rationale
POS & Store Systems	On-Premises	ON-PREM	Latency-sensitive store ops; direct hardware integration with POS terminals and barcode scanners
Inventory Management	On-Premises	ON-PREM	Integrated with warehouse WMS infrastructure; regulated stock data kept within facility boundaries
ERP / Legacy Order Core	On-Premises	ON-PREM	Deep integration with financial ledger and procurement systems; cannot tolerate migration disruption
Order Processing Service	Hybrid	HYBRID	Routes cloud e-commerce orders to nearest fulfillment center; bridges cloud API tier and on-prem ERP
Inventory Sync Service	Hybrid	HYBRID	Publishes real-time stock events to cloud; subscribes to warehouse updates from on-prem systems
Product Microservice	Cloud — IBM IKS	CLOUD	Stateless, horizontally scalable; supports 50K concurrent users during peak; no regulated data stored

Component	Environment	Placement	Rationale
Customer Profile Service	Cloud — IBM IKS	CLOUD	Scales independently; EU customer data isolated to IBM Cloud Frankfurt (GDPR-compliant region)
Checkout & Payment API	Cloud — IBM IKS	CLOUD	PCI-DSS compliant; isolated in dedicated namespace; integrates with external payment processors via TLS
API Gateway	Cloud — IBM IKS	CLOUD	Centralized ingress for all partner and storefront traffic; rate limiting, auth, and routing enforced here
CDN / Static Assets	Cloud — IBM CIS	CLOUD	Global edge caching for storefront assets; reduces origin load; low-latency delivery to all regions
CI/CD Pipeline	Cloud — IBM Toolchain	CLOUD	Automated build, test, security scan, and deployment; targets Dev / Staging / Prod IKS clusters
Observability Platform	Cloud — IBM Cloud	CLOUD	Centralized metrics, traces, and logs from all services; dashboards for 100% critical path coverage
Customer PII Database (EU)	On-Premises / Cloud FR	ON-PREM	GDPR-mandated data residency; EU customer records stored exclusively within Frankfurt region or on-prem EU DC
Payment Token Vault	On-Premises	ON-PREM	PCI-DSS scope isolation; payment tokens stored on-prem behind hardware security modules (HSMs)
Disaster Recovery	Cloud — IBM Cloud	CLOUD	Cross-region replication to meet RTO 4h / RPO 1h objectives; automated failover orchestration

3.1 Placement Decision Principles

The following principles govern placement of workloads not explicitly listed above:

- Any service that processes, stores, or transmits EU customer personal data must reside in either the IBM Cloud Frankfurt region or the on-premises EU data center — not in US-South or any other region.
- Any service within PCI-DSS scope must be deployed in the dedicated Payment Isolation Zone and must not share network segments with services outside that scope.
- Stateless, horizontally scalable services with no regulated data dependencies are default-cloud — they benefit from IKS auto-scaling and should not be constrained to on-premises capacity.

- Legacy systems with direct database or hardware dependencies (ERP financial ledger, warehouse WMS barcode scanners, POS terminals) remain on-premises until a formal re-platforming project is scoped and approved.

4. Connectivity Design

Secure, low-latency connectivity between the on-premises data centers and IBM Cloud is the critical path for the hybrid architecture. All on-premises-to-cloud traffic traverses a dedicated Hybrid Connectivity Zone (10.0.40.0/24) with strict access control lists permitting only approved traffic types. The table below documents the full connectivity model.

Connectivity Method	Type	Mechanism	Justification
IBM Cloud Direct Link 2.0	Primary hybrid link	10 Gbps dedicated private circuit between on-prem DC and IBM Cloud; no public internet traversal	Meets latency SLA (<2s); required for real-time inventory sync and order routing across hybrid boundary
IPSec VPN (Backup)	Secondary / failover	Encrypted site-to-site VPN tunnel over internet; activates automatically if Direct Link degrades	Ensures 99.95% availability target is not compromised by a single connectivity failure
IBM Transit Gateway	Cloud-internal routing	Connects multiple IBM Cloud VPCs and regional environments through a single managed hub	Enables EU (Frankfurt) and US (Dallas) cloud regions to share internal routes without traversing public internet
TLS 1.3 API Channels	Partner integrations	All third-party logistics providers and payment processors connect via HTTPS/TLS 1.3 over public internet with mTLS certificate pinning	PCI-DSS and GDPR require encryption in transit; mTLS ensures only authorized partners can establish sessions
IBM Satellite Link	On-prem management	Extends IBM Cloud control	Allows hybrid workloads (order processing, inventory sync) to be

Connectivity Method	Type	Mechanism	Justification
		plane capabilities to on-premises clusters; enables unified workload management and policy enforcement	managed through the same toolchain as cloud-native services

4.1 Primary Connectivity — IBM Cloud Direct Link 2.0

IBM Cloud Direct Link 2.0 is the primary hybrid connectivity mechanism. A dedicated 10 Gbps private circuit connects each on-premises data center directly to the IBM Cloud backbone, bypassing the public internet entirely. This is the architectural requirement that enables the real-time inventory synchronization objective (FR-01: updates propagated within 5 seconds) and ensures that the order routing service can communicate with the on-premises ERP with latency consistent with the 2-second end-to-end transaction SLA.

Why Direct Link Over VPN?

- Consistent, guaranteed bandwidth — no contention with general internet traffic during Black Friday peak loads
- Lower and more predictable latency — dedicated circuit eliminates jitter caused by shared internet paths
- No encryption overhead on the primary path — Direct Link traffic is private by design; IPSec VPN adds CPU overhead that accumulates at high throughput
- IBM SLA-backed availability — Direct Link carries an IBM uptime SLA; internet-based VPN does not

4.2 Failover Connectivity — IPSec VPN

An IPSec VPN tunnel is maintained in standby mode at all times and activates automatically within 30 seconds if the Direct Link circuit degrades or fails. The VPN is configured with IKEv2 and AES-256-GCM encryption, consistent with the NFR-04 encryption standard. While VPN throughput (typically 500 Mbps–1 Gbps) is lower than Direct Link, it is sufficient to maintain inventory sync and order routing during the failover window. The combination of Direct Link primary and VPN failover supports the 99.95% availability SLA even during carrier-level outages.

5. Network Segmentation

The hybrid network is divided into nine distinct segments, each serving a specific functional and security purpose. Segmentation prevents lateral movement between tiers, enforces the principle of least privilege, and isolates regulated workloads — particularly PCI-DSS payment systems and GDPR-

governed EU data — from general application traffic. All inter-segment traffic is subject to explicit firewall allow-list rules; default-deny applies to all other flows.

Network Segment	CIDR / Range	Primary Components	Security Controls & Purpose
Public DMZ	10.0.1.0/24	IBM Cloud Internet Services (CIS), CDN, Global Load Balancer	First line of ingress for all customer and partner traffic; DDoS protection and WAF enforced at this tier; no direct backend access
API Gateway Zone	10.0.2.0/24	IBM API Connect / Kong Gateway on IKS ingress	Authenticates all API consumers; enforces rate limits, JWT validation, and TLS termination before traffic reaches application services
Application Services Zone	10.0.10.0/22	Product, Order, Customer, Checkout microservices on IKS	Internal east-west service mesh (Istio); mTLS between all pods; no direct inbound from public internet; accessible only via API Gateway
Payment Isolation Zone	10.0.20.0/24	Checkout & Payment API — dedicated IKS namespace	PCI-DSS requires hard isolation of cardholder data; separate namespace with restrictive NetworkPolicy; egress only to approved payment processors
Data Persistence Zone	10.0.30.0/24	IBM Cloud Databases (PostgreSQL, Redis), Object Storage	No inbound from internet or API Gateway; services connect via private service endpoint; encrypted at rest (AES-256)
Hybrid Connectivity Zone	10.0.40.0/24	Direct Link termination, VPN gateway, IBM Transit Gateway	Dedicated subnet for on-prem-to-cloud traffic; strict ACLs permit only inventory sync, order routing, and management traffic
On-Premises — Internal LAN	192.168.0.0/16	ERP, Inventory WMS, POS systems, PII DB, HSM vault	Air-gapped from public internet; communicates with cloud exclusively through Hybrid Connectivity Zone; firewall enforces allowlist
Management & Observability	10.0.50.0/24	IBM Cloud Monitoring, Log Analysis, Sysdig, CI/CD agents	Separate from application traffic; only authorized DevOps and SRE roles have access; audit logs shipped to immutable object storage
EU Compliance Zone	10.1.0.0/16 (FRA)	Customer Profile DB (Frankfurt region)	GDPR-mandated data residency; EU customer PII is processed and stored exclusively within this zone; cross-region transfer blocked by policy

5.1 Key Segmentation Design Decisions

Payment Isolation Zone: The Checkout and Payment API is deployed in a dedicated Kubernetes namespace with a strict NetworkPolicy that restricts all inbound traffic to connections from the API Gateway and all outbound traffic to the approved list of payment processor endpoints. This hard boundary defines the PCI-DSS cardholder data environment (CDE) within the cloud. No logging infrastructure that operates outside the PCI scope can access pod-level network traffic within this zone.

EU Compliance Zone: All IBM Cloud resources that process or store EU customer PII are provisioned exclusively in the Frankfurt region under a dedicated IBM Cloud account sub-scope. IBM Cloud Identity and Access Management (IAM) policies block any cross-region data transfer API calls from this zone. This architectural control — not a policy document — is what satisfies the GDPR data residency obligation raised by the Security and Compliance Director.

Istio Service Mesh for East-West Security: Within the Application Services Zone, Istio enforces mutual TLS (mTLS) for all pod-to-pod communication. This means that even if an attacker compromises one microservice, they cannot communicate with other services without a valid client certificate. The service mesh also enforces traffic policies — a single misconfigured service cannot create an unintended network path to the Data Persistence Zone.

6. Traffic Distribution

The platform employs a layered traffic distribution model that applies routing intelligence at each level of the stack — from global DNS through to individual pod selection. This approach supports the peak demand requirement of 50,000 concurrent users, the 99.95% availability target, and the sub-2-second transaction latency SLA simultaneously.

Traffic Distribution Component	Layer	Mechanism	Platform Benefit
IBM Cloud Internet Services (CIS)	Layer 7 — Global	Anycast DNS + global load balancer routes users to closest healthy IBM Cloud region; health checks every 30s	Handles geographic routing during normal ops and DNS failover within 60s if a regional endpoint fails — directly supports 99.95% availability SLA
IBM Cloud Application Load Balancer	Layer 7 — Regional	HTTP/HTTPS-aware routing; path-based rules direct /products, /orders, /checkout to their respective service	Application-aware routing isolates service blast radius; enables blue-green and canary deployments without DNS changes

Traffic Distribution Component	Layer	Mechanism	Platform Benefit
		namespaces on IKS	
Kubernetes Ingress (NGINX)	Layer 7 — Cluster	Terminates TLS inside the cluster; routes to ClusterIP services based on Host header and URL path; enforces pod-level NetworkPolicy	Provides fine-grained routing within IKS; integrates with cert-manager for automated TLS certificate renewal
Istio Service Mesh	Layer 7 — East-West	mTLS for all pod-to-pod communication; traffic policies, circuit breakers, and retry logic configured per service; distributed tracing enabled	Eliminates unencrypted internal traffic; circuit breakers prevent cascading failures during peak load (Black Friday scenarios)
IBM Cloud Network Load Balancer	Layer 4 — TCP/UDP	High-throughput TCP load balancing for Direct Link and VPN gateway traffic between on-prem and cloud hybrid zone	Low-overhead routing for bulk inventory sync and ERP data streams that do not require HTTP-level inspection
DNS Failover (CIS Health Checks)	DNS — Failover	If primary US-South region health check fails, CIS automatically updates DNS to route traffic to EU-Frankfurt standby within TTL window	Supports RTO 4h objective; automated failover removes manual intervention from regional outage response

6.1 Peak Load Strategy

During high-traffic events such as Black Friday or seasonal sales campaigns, the following traffic distribution capabilities activate in combination:

- IBM CIS global load balancer distributes new connections across US-South and EU-Frankfurt regions based on real-time health scores and geographic proximity

- IKS Horizontal Pod Autoscaler (HPA) scales Product and Checkout service pod counts from baseline 3 replicas to up to 20 replicas within 90 seconds of sustained CPU/RPS threshold breach
- IBM Cloud Block Storage IOPS burst capacity absorbs the increased database write load from concurrent checkout transactions
- Redis in-memory cache absorbs product catalogue and session read traffic — preventing read amplification from propagating to the PostgreSQL transactional layer
- Istio circuit breakers automatically shed load from degraded upstream dependencies rather than allowing timeout queues to cascade across the service mesh

7. Storage Strategy

The storage model is designed around the access pattern, compliance classification, and durability requirements of each data type. The Global Retail Platform uses four storage models — block, object, file, and in-memory — each selected to match the performance and regulatory profile of its data.

Data Type	Storage Model	Technology	Location	Justification
Customer transaction records	Block Storage	IBM Cloud Block Storage (10 IOPS/GB tier)	On-Premises / Cloud DB Zone	Low-latency random read/write required by transactional RDBMS (PostgreSQL); consistent IOPS critical for <2s checkout latency SLA
Inventory & order state	Block Storage	IBM Cloud Block Storage (5 IOPS/GB tier)	On-Premises + Hybrid DB	ACID-compliant storage for inventory WMS; replicated snapshot to cloud every 15min to meet RPO 1h objective
EU Customer PII	Block Storage (encrypted)	IBM Cloud Block Storage — Frankfurt region only	Cloud EU Zone (GDPR)	Encrypted at rest (AES-256); stored exclusively in Frankfurt to satisfy GDPR data residency; access logged to immutable audit trail
Payment token vault	Block Storage + HSM	On-premises HSM-backed encrypted block volumes	On-Premises (PCI scope)	Hardware security module enforces PCI-DSS key management; tokens never leave on-prem PCI-compliant boundary

Data Type	Storage Model	Technology	Location	Justification
Product catalogue & media assets	Object Storage	IBM Cloud Object Storage (Standard tier)	Cloud — Multi-region	Ideal for large, unstructured binary objects; geo-redundant buckets serve CDN edge nodes; no block I/O overhead for static content
Build artifacts, SBOM, scan reports	Object Storage	IBM Cloud Object Storage (Archive tier)	Cloud — CI/CD Zone	Cost-efficient long-term retention of audit evidence; immutable retention policy supports compliance obligations; auto-tiered to cold storage after 90 days
Application and pipeline logs	Object Storage	IBM Cloud Object Storage (Smart Tier)	Cloud — Observability Zone	Append-only audit logs stored in immutable COS bucket; ingested by IBM Log Analysis; access requires MFA and IAM role
Shared configuration & feature flags	File Storage	IBM Cloud File Storage (NFS, Endurance tier)	Cloud — App Services Zone	ReadWriteMany access mode required; multiple microservice pods on IKS read shared config files without duplication; low-latency NFS mounts
CI/CD shared workspace	File Storage	IBM Cloud File Storage — pipeline scratch volumes	Cloud — Toolchain Zone	Build agents require shared persistent volumes across pipeline stages for passing build.env and test artifacts between steps
In-memory session cache	In-Memory / Cache	IBM Databases for Redis (HA cluster)	Cloud — App Services Zone	Sub-millisecond read latency for session tokens and product catalogue hot data; supports 50K concurrent users without DB read amplification
Disaster recovery snapshots	Object Storage	IBM COS cross-region	Cloud — DR Zone	Automated nightly snapshots replicated cross-region; point-

Data Type	Storage Model	Technology	Location	Justification
	(cross-region)	replication to secondary DC		in-time restore supports RPO 1h; DR runbook tested quarterly against RTO 4h

7.1 Storage Design Rationale

Block Storage for Transactional Data: Customer transactions, inventory records, and order state require ACID-compliant relational databases (PostgreSQL). Block storage provides the consistent, low-latency IOPS that transactional workloads require. Object storage, while cheaper, introduces latency variability that is incompatible with the <2-second end-to-end transaction SLA. IBM Cloud Block Storage at the 10 IOPS/GB performance tier is provisioned for the checkout database to sustain peak concurrent write throughput.

Object Storage for Assets and Audit Evidence: Product catalogue images, build artifacts, SBOM files, security scan reports, and application logs do not require the random-access performance of block storage. IBM Cloud Object Storage with smart tiering provides cost-efficient, geo-redundant storage with immutable retention policies — a direct requirement for compliance audit evidence. The immutability feature ensures that once a security scan report or deployment log is written, it cannot be altered or deleted, satisfying audit traceability requirements for both GDPR and PCI-DSS.

File Storage for Shared Operational Data: Shared configuration files, feature flag manifests, and CI/CD pipeline workspace volumes require ReadWriteMany (RWX) access — multiple pods must read and occasionally write to the same mounted volume simultaneously. NFS-backed IBM Cloud File Storage supports the RWX access mode that Kubernetes block volumes do not, making it the only viable choice for these workloads. Block volumes are ReadWriteOnce (RWO), which would force a single-pod bottleneck on shared configuration access.

In-Memory Cache for Hot Data: IBM Databases for Redis serves as the hot data tier for product catalogue reads and session tokens. During a peak demand event, the product listing endpoint can receive thousands of requests per second. Without a cache layer, each request would execute a full database read, creating a read amplification pattern that would overwhelm the PostgreSQL cluster and push transaction latency beyond the 2-second SLA. Redis delivers sub-millisecond read responses and absorbs the majority of catalogue read traffic before it reaches the database tier.

8. Stakeholder Requirements Traceability

The architecture decisions documented in this paper trace directly to the stakeholder requirements established in the requirements analysis phase. The table below maps each stakeholder's key concern to the specific architectural controls that address it.

Stakeholder	Key Requirement	How the Architecture Addresses It
VP, E-Commerce	Platform stability during peak events; <2s checkout latency; 50K concurrent users	IKS auto-scaling + Istio circuit breakers absorb Black Friday spikes; rolling update strategy with automatic rollback eliminates planned downtime windows; CIS global load balancer provides sub-60s regional failover
CTO	Cloud-native architecture enabling rapid feature delivery	Microservices on IKS with independent CI/CD pipelines (IBM Toolchain) allow per-service deployments; IBM Satellite provides unified hybrid management without re-architecting on-prem systems
Head of Supply Chain	Real-time inventory sync across 12 warehouses within 5 seconds	Inventory Sync Service publishes stock events via IBM Event Streams (Kafka); Direct Link ensures <10ms hybrid latency; on-prem WMS integration preserved through Hybrid zone
Security & Compliance Director	GDPR, PCI-DSS, data residency, encryption at rest and in transit	EU Compliance Zone (Frankfurt) enforces GDPR residency; Payment Isolation Zone (dedicated namespace + HSM) meets PCI-DSS; AES-256 at rest + TLS 1.3 in transit applied universally; immutable audit logs in COS
DevOps Engineering Lead	Automated CI/CD; eliminate manual release bottlenecks; fast MTTR	7-stage IBM Toolchain pipeline (Build→Test→Security→Artifact→Dev→Staging→Prod) with automated security gates; kubectl rollout undo enables <5min rollback; centralized observability reduces MTTR

9. Compliance and Regulatory Controls

The hybrid architecture is designed to satisfy all regulatory and non-functional requirements identified in the requirements document. The following table maps each compliance obligation to its controlling architectural mechanism.

Requirement	Obligation	Architectural Control	Implementation Detail
GDPR	EU customer PII data residency and access control	IBM Cloud Frankfurt (EU) region + on-prem EU DC; EU Compliance Zone with blocked cross-region transfer	AES-256 encryption at rest; IAM access controls; immutable audit logs in COS; DSAR support via Customer Profile Service API
PCI-DSS	Cardholder data isolation and key management	Dedicated Payment Isolation Zone (IKS namespace); payment tokens stored in on-prem HSM vault	TLS 1.3 in transit; HSM-backed key management; network segmentation; quarterly penetration testing; no card data in cloud logs
Availability (NFR-01)	99.95% uptime (≤4.4 hrs downtime/month)	Multi-region active-standby; CIS global load balancer with DNS	Istio circuit breakers; automated rollback on health check failure; rolling updates with maxUnavailable=0

Requirement	Obligation	Architectural Control	Implementation Detail
		failover; IKS multi-zone worker pools	
Resilience (NFR-03)	RTO ≤ 4 hours; RPO ≤ 1 hour	Cross-region DR replication in IBM COS; automated failover orchestration via IBM Satellite	15-minute snapshot intervals for inventory/order state; quarterly DR drills against RTO/RPO targets
Encryption (NFR-04)	TLS 1.3 in transit; AES-256 at rest	Enforced universally across all cloud and hybrid zones; no plaintext data paths permitted	Istio mTLS for internal east-west traffic; cert-manager for automated certificate lifecycle; IBM Key Protect for key management

10. Architecture Summary

The hybrid cloud architecture designed for the Global Retail Platform addresses the core tension of the modernization initiative: the organization must scale to meet e-commerce growth and peak demand while simultaneously honoring regulatory constraints that prevent certain data from leaving defined geographic and infrastructure boundaries.

The architecture resolves this tension through deliberate workload placement — cloud-native, stateless services handle the scalable customer-facing workload on IBM IKS in US-South, while regulated data (EU PII, payment tokens) remains within GDPR and PCI-DSS mandated boundaries. IBM Cloud Direct Link 2.0 provides the private, low-latency backbone that makes real-time inventory synchronization and hybrid order routing possible without exposing internal systems to the public internet.

Network segmentation enforces security boundaries that cannot be bypassed by misconfiguration — the Payment Isolation Zone and EU Compliance Zone are architectural controls, not policy documents. The Istio service mesh extends that security posture to east-west internal traffic. The layered traffic distribution model — from CIS global load balancing at the DNS layer to Istio circuit breakers at the pod level — ensures that the platform can absorb 50,000 concurrent users during Black Friday without a manual intervention.

The storage model aligns each data type to the access pattern and durability profile that it requires, with immutable object storage providing the audit trail that both GDPR and PCI-DSS compliance programs depend on. Taken together, these design decisions produce a platform that is simultaneously scalable, secure, compliant, and operationally manageable — satisfying the objectives of every stakeholder identified in the requirements phase.